

CRYPTOLOGY

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Lab Assignment

Importance of Cryptography: Using Wireshark, investigate the transmission of dummy login credentials over HTTP and HTTPS. Analyse the captured network traffic, compare how the credentials are transmitted in both protocols, and explain the role of encryption in protecting sensitive information.

Answer:

Introduction

In this practical, Wireshark is used to see how login information travels across a network. The aim is to compare HTTP and HTTPS and understand how encryption helps protect sensitive information such as usernames and passwords.

Procedure

1: Starting Packet Capture

First, I opened Wireshark in Kali Linux and selected the eth0 network interface. Then I started capturing packets so that all network traffic could be recorded.

Step 2: Testing HTTP Login

I opened the following HTTP login page in Firefox: <http://test/login>

The page contained a username field and a password field. I entered dummy credentials and clicked the Login button. After submitting the form, Wireshark captured the packets generated by the browser.

Step 4: Analysing HTTPS Traffic

Unlike the HTTP traffic, the HTTPS packets showed **TLS 1.2** and **Encrypted Application Data**. Although details such as IP addresses and port numbers could still be seen, the actual login information was not visible because it was encrypted.

HTTP

- Connection was not secure.
- Data was transmitted without encryption.
- Sensitive information could be exposed if someone captures the traffic.

HTTPS

- Connection used TLS encryption.
- Wireshark displayed the data as Encrypted Application Data.
- Login information could not be viewed directly.

Feature	HTTP	HTTPS
Encryption	No	Yes
Port Number	80	443
Security Level	Low	High
Data Visibility	Can be viewed	Encrypted
Suitable for Login Credentials	No	Yes

Importance

clearly showed why cryptography is important. In HTTP, information travels without protection, making it easier for attackers to view sensitive data if they capture network traffic.

HTTPS solves this problem by encrypting the data before sending it across the network. Even if someone captures the packets, they cannot easily read the actual information.

Conclusion

From this practical, I learned that HTTP and HTTPS handle data very differently. HTTP does not protect information during transmission, while HTTPS uses TLS encryption to secure the communication.

The Wireshark captures showed that HTTPS hides sensitive information by encrypting it, making it much safer for login pages and other websites that handle personal data. Therefore, HTTPS should always be used when transmitting usernames, passwords, or any confidential information.